

ENTHUSIAST ADVANCE COURSE

PHASE : MEA & MEPS-I (A)

TARGET : PRE MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 16-07-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	3	2	4	3	2	1	2	2	3	2	1	4	2	3	4	4	2	3	4	2	4	4	2	2	3	3	2	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	2	1	4	3	4	2	2	2	1	2	1	1	3	4	3	1	1	1	2	4	1	4	2	2	2	2	3	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	2	4	2	1	1	2	2	2	4	1	1	2	2	1	1	3	2	3	2	1	4	3	3	3	1	4	2	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	1	1	4	1	3	1	2	2	4	4	4	2	2	3	3	2	3	4	4	3	4	4	4	1	4	3	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	3	2	2	3	3	4	1	4	2	3	4	2	3	3	4	3	3	4	3	1	3	1	4	4	4	2	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	3	4	4	3	3	4	4	4	2	4	3	4	3	3	4	3	3	4	3	4	3	4	2	3	3	2	1	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	2	4	3	1	4	3	3	4	1	2	2	3	4	4	3	4	3	4	1										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. **Ans (3)**

$$f_1 - f_2 = 2 \Rightarrow v \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) = 2$$

$$v \left(\frac{1}{2} - \frac{1}{2.02} \right) = 2$$

$$v = \frac{2 \times 2 \times 2.02}{0.02} = 404 \text{ m/s}$$

2. **Ans (3)**

Answer (3)

$$k = \frac{2\pi}{\lambda} = \frac{\pi}{10} \Rightarrow \lambda = 20 \text{ cm}$$

$x = 0$ is antinode $\therefore 2^{\text{nd}}$ antinode at $\frac{\lambda}{2} = 10$ cm, 3^{rd} antinode, will be at $x = \lambda = 20$ cm

3. **Ans (2)**

$$\lambda = 2(\ell_2 - \ell_1) = 2(45 - 15) = 66 \text{ cm}$$

$$v = f\lambda = 500 \times \frac{66}{100} = 330 \text{ m/s}$$

4. **Ans (4)**

$$n = \frac{1}{2\ell} \sqrt{\frac{T}{\pi r^2 \rho}}, \quad m = \frac{M}{\ell}$$

$$n \propto \frac{1}{\ell r} \Rightarrow \frac{n_1}{n_2} = \frac{r_2}{r_1} \times \frac{\ell_2}{\ell_1} = \frac{r}{2r} \times \frac{2L}{L} = 1$$

5. **Ans (3)**

Answer (3)

$$\frac{2x}{v} = 4$$

$$x = 2v \quad \dots(i)$$

$$\frac{2(x - 100)}{v} = 3.5$$

$$x - 100 = \frac{v \times 7}{4} \quad \dots(ii)$$

$$100 = 2v - \frac{7v}{4}$$

$$100 = \frac{v}{4}$$

$$v = 400 \text{ m/s}$$

6. **Ans (2)**

$$\Rightarrow f_{\text{app}} = f_0 \left(\frac{v + v_0}{v} \right)$$

$$\Rightarrow \frac{f_{\text{app}} - f_0}{f_0} = \frac{v_0}{v} = \frac{\frac{v}{5}}{v} = \frac{1}{5} = 20\%$$

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7. Ans (1)

$$f' = f \left(\frac{v + v_x}{v - v_x} \right) = 1000 \left(\frac{330 + 220}{330 - 220} \right) = 5000 \text{ Hz}$$

8. Ans (2)

$$\Delta\lambda = \frac{v}{c} \lambda = \frac{10^6}{3 \times 10^8} \times 5400 = 18 \text{ \AA}$$

9. Ans (2)

$$\begin{aligned} n &= 5 \times 10^{24} \times P \times \lambda \\ &= 5 \times 10^{24} \times 2 \times 10^{-3} \times \frac{C}{v} \\ &= 5 \times 10^{24} \times 2 \times 10^{-3} \times \frac{3 \times 10^8}{6 \times 10^{14}} \\ n &= 5 \times 10^{15} \end{aligned}$$

10. Ans (3)

Let p be the momentum of a photon arriving at the mirror, we then have $p = \frac{h}{\lambda}$

The change in momentum for a photon

$$= p - (-p) = 2p = \frac{2h}{\lambda}$$

Force = rate of change of momentum of the photons = $n \left(\frac{2h}{\lambda} \right)$; Where n is the rate at which the photons hitting the mirror.

11. Ans (2)

$$h\nu = \phi + e\nu_0 \quad \dots(1)$$

$$2h\nu = \phi + e\nu_0' \quad \dots(2)$$

$$(1) \times (2)$$

$$2h\nu = 2\phi + 2e\nu_0 \quad \dots(3)$$

By (2) & (3)

$$e\nu_0' = 2e\nu_0 + \phi$$

$$\nu_0' = 2\nu_0 + \frac{\phi}{e}$$

$$\nu_0' > 2\nu_0 \text{ more than double}$$

12. Ans (1)

$$\begin{aligned} \frac{1}{2}mv^2 &= k_{\max} = E - \phi \\ \frac{1}{2}m(6 \times 10^6)^2 &= 3\phi - \phi = 2\phi \quad \dots(1) \end{aligned}$$

$$\frac{1}{2}mv_2^2 = 9\phi - \phi = 8\phi \quad \dots(2)$$

divide [equation (2), (1)]

$$\begin{aligned} \frac{v_2^2}{(6 \times 10^6)^2} &= 4 \\ v_2 &= 12 \times 10^6 \end{aligned}$$

13. Ans (4)

According to Einstein equation $h\nu = h\nu_0 + K_{\max}$

$$\Rightarrow K_{\max} = h\nu - h\nu_0$$

on comparing it with $y = mx + c$ it is clear to say that,

This is the equation of straight line having positive slope (h) and negative intercept ($h\nu_0$) on KE axis.

16. Ans (4)

$$\text{Least count} = \frac{\text{Pitch}}{\text{Number of division circular scale}}$$

$$= \frac{1 \text{ mm}}{200} = 0.005 \text{ mm}$$

$$\text{Reading} = 5 \text{ mm} + (50)(0.005)$$

$$= 5 \text{ mm} + 0.250 \text{ mm}$$

$$= 5.250 \text{ mm}$$

18. Ans (2)

$$\text{Least count of vernier callipers} =$$

$$\left(1 - \frac{9}{10} \right) \times 1 \text{ mm} = 0.1 \text{ mm}$$

$$\text{Zero error} = \text{VSR} \times \text{LC} = 3 \times 0.1 = 0.3 \text{ mm}$$

$$\text{Reading} = \text{MSR} + \text{VSR} \times \text{LC} = 13 \text{ mm} + 7 \times 0.1 \text{ mm} = 13.7 \text{ mm}$$

$$\text{Correct reading} = \text{Reading} - \text{Zero error} = 13.7 - 0.3 \text{ mm} = 13.4 \text{ mm}$$

19. Ans (3)

$$Q = CV$$

$$C = \frac{\epsilon_0 A}{d} \Rightarrow C \downarrow$$

$$\Rightarrow Q \downarrow, V = \text{constant}$$

$$U = \frac{1}{2} CV^2 \downarrow$$

24. Ans (2)

At the mid-point of the line joining the conductors, $\vec{B} = 0$. As we come close to the wires, the magnitude of B increase. The direction of magnitude fields on opposite sides of a wire will be opposite. Again,

$$\vec{B} = 0, \text{ as } r \rightarrow \infty$$

32. Ans (2)

For power to be maximum

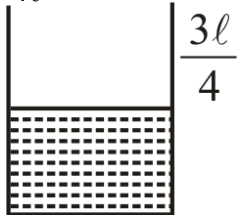
External resistance = Equivalent internal resistance of the circuit

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SECTION-B

36. Ans (3)

$$\frac{V}{4\ell} = 220 \text{ Hz}$$



$$n' = 3 \frac{V}{4 \times \frac{3\ell}{4}} = \frac{V}{\ell} = 880 \text{ Hz}$$

37. Ans (4)

$$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{4 \times 10^{10}}{1600}} = 5 \times 10^3 \text{ m/s}$$

$$f = \frac{v}{2\ell} = \frac{5 \times 10^3}{2 \times 20} \times 100 = 12.5 \text{ kHz}$$

38. Ans (2)

$$p = \frac{h}{\lambda} = \frac{6.62 \times 10^{-34}}{10^{-10}} = 6.62 \times 10^{-24} \text{ kgm/sec}$$

39. Ans (2)

$$4.8 = \frac{hc}{e\lambda} - \frac{hc}{e\lambda_0} \quad \dots(1)$$

$$1.6 = \frac{hc}{e(2\lambda)} - \frac{hc}{e\lambda_0} \quad \dots(2)$$

$$\frac{(1)}{(2)} \quad 3 = \frac{\frac{hc}{e\lambda} - \frac{hc}{e\lambda_0}}{\frac{hc}{2e\lambda} - \frac{hc}{e\lambda_0}}$$

$$\frac{(1)}{(2)} \quad 3 = \frac{\frac{hc}{e\lambda} - \frac{hc}{e\lambda_0}}{\frac{hc}{2e\lambda} - \frac{hc}{e\lambda_0}}$$

on solving we get $\lambda_0 = 4\lambda$

40. Ans (2)

In electric field photoelectron will experience force and accelerate opposite to the field so its K.E. increase (i.e., stopping potential will increase), no change in photoelectric current, and threshold wavelength.

41. Ans (1)

Spring constant

$$K = [M^1 L^0 T^{-2}]$$

$$\text{pressure} = [M^1 L^{-1} T^{-2}]$$

$$\text{frequency} = [M^0 L^0 T^{-1}]$$

$$\text{work} = [M^1 L^2 T^{-2}]$$

42. Ans (2)

$$\text{Error} = \frac{\Delta t}{t} \times 100 = \frac{1}{5 \times 25} \times 100 = 0.8\%$$

43. Ans (1)

$$n_1 u_1 = n_2 u_2$$

$$n_1 = n_2 = g$$

$$\frac{L_1}{L_2} = \left(\frac{T_1}{T_2} \right)^2 = 4$$

46. Ans (4)

$$q_i = C_i V = 2V = q$$

This charge will remain constant after switch S is shifted from position 1 to position 2.

$$U_i = \frac{1}{2} \frac{q^2}{C_i} = \frac{q^2}{2 \times 2} = \frac{q^2}{4}$$

$$U_f = \frac{1}{2} \frac{q^2}{C_f} = \frac{q^2}{2 \times 10} = \frac{q^2}{20}$$

$$\text{Energy dissipated} = U_i - U_f = \frac{q^2}{5}$$

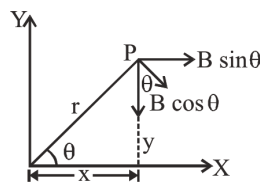
The energy dissipated is 80% of the initial stored energy as % decrease in energy

$$= \frac{U_i - U_f}{U_i} \times 100\% = \frac{q^2/5}{q^2/4} \times 100\%$$

$$= \frac{4}{5} \times 100\% = 80\%$$

48. Ans (1)

Magnetic field at P is $\vec{B}$, perpendicular to OP in the direction shown in figure.



$$\text{So, } \vec{B} = B \sin \theta \hat{i} - B \cos \theta \hat{j}$$

$$\text{Here } B = \frac{\mu_0 I}{2\pi r}$$

$$\sin \theta = \frac{y}{r} \text{ and } \cos \theta = \frac{x}{r}$$

$$\therefore \vec{B} = \frac{\mu_0 I}{2\pi} \cdot \frac{1}{r^2} (y\hat{i} - x\hat{j}) = \frac{\mu_0 I (y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)} \quad (\text{as } r^2 = x^2 + y^2)$$

50. Ans (1)

$$\text{Since } P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$$

Resistance is inversely proportional to power for a constant voltage.

So, 850 W setting has higher resistance.

$$R = \frac{V^2}{P} = \frac{(120)^2}{850} = 17\Omega$$

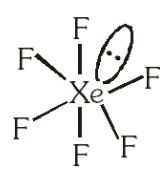
$$R = \frac{V^2}{P} = \frac{(120)^2}{1250} = 12\Omega$$

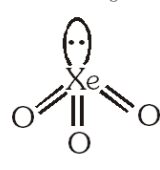
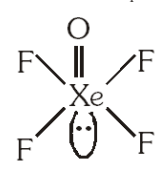
SUBJECT : CHEMISTRY

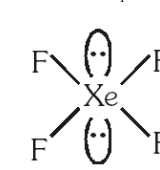
SECTION-A

57. **Ans (2)**
 Amphoterics + NaOH $\rightarrow$ H₂
60. **Ans (1)**
 $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HOCl}$
69. **Ans (2)**
 For ion-dipole interaction : $E \propto \frac{1}{r^2}$
71. **Ans (1)**
 Due to the symmetrical structure dipole moment of (3) & (4) are zero & (1) having maximum dipole moment.

SECTION-B

86. **Ans (1)**
- XeF_6

 distorted octahedral

XeO_3

 Pyramidal
- XeOF_4

 Square pyramidal

XeF_4

 Square planar

87. **Ans (4)**
 NCERT-XII, f-Block
91. **Ans (1)**
 Conceptual and fact based

SUBJECT : BIOLOGY-I

SECTION-A

102. **Ans (4)**
 NCERT-XI Pg#215

103. **Ans (4)**
 NCERT-XI Pg#208
104. **Ans (2)**
 NCERT-XI Pg#211
105. **Ans (2)**
 NCERT-XI Pg#212
107. **Ans (3)**
 NCERT-XI Pg#214
108. **Ans (2)**
 NCERT-XI Pg#212
109. **Ans (3)**
 NCERT-XI Pg#217
110. **Ans (4)**
 NCERT-XI Pg#208
113. **Ans (4)**
 NCERT-XI Pg#216
114. **Ans (4)**
 NCERT-XI Pg#217
116. **Ans (1)**
 NCERT-XI Pg#213
117. **Ans (4)**
 NCERT-XI Pg#224
118. **Ans (3)**
 NCERT XI Pg. # 222, fig. 13.10
119. **Ans (2)**
 NCERT XI Pg. # 92, Figure 6.7(b)
120. **Ans (3)**
 NCERT XIth Pg#85, Fig-6.1(a,b)
121. **Ans (3)**
 NCERT (XIth) Pg # 85-6.1.1
122. **Ans (3)**
 NCERT (XIth) Pg # 88-6.1.2.2
123. **Ans (3)**
 NCERT (XIth) Pg # 94-6.3.6
124. **Ans (2)**
 NCERT Pg. # 86 and 87

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125. **Ans (2)**

NCERT (XIth) Pg. # 86

126. **Ans (3)**

NCERT (XIth) Pg. # 89

127. **Ans (3)**

NCERT-XI Page No. 90

128. **Ans (4)**

NCERT (XI) Pg. # 90

129. **Ans (1)**

NCERT XI, Page # 80, Para - 5.9.2

130. **Ans (4)**

NCERT Pg. # 79

131. **Ans (2)**

NCERT XIth Pg. # 79, 5.9.2

132. **Ans (3)**

NCERT XI, Pg.# 79

134. **Ans (2)**

NCERT XII Page No. # 157 (E), 173 (H)

135. **Ans (3)**

NCERT (XIIth) Pg. # 155

SECTION-B

137. **Ans (4)**

NCERT-XI Pg#211, 212

138. **Ans (3)**

NCERT-XI Pg#216

140. **Ans (4)**

NCERT-XI Pg#213

141. **Ans (3)**

NCERT-XI Pg#218

142. **Ans (1)**

NCERT-XI Pg#219

143. **Ans (3)**

NCERT XI pg.# 85

144. **Ans (1)**

NCERT XI, Pg. # 91(E), 90-91(H)

145. **Ans (4)**

NCERT XI, Pg. # 88, 89

146. **Ans (4)**

NCERT XII Pg.# 86

147. **Ans (4)**

NCERT (XI) Pg. # 81

148. **Ans (2)**

NCERT (XI) Pg # 79

SUBJECT : BIOLOGY-II

SECTION-A

152. **Ans (3)**

NCERT (XII) Pg. # 158

155. **Ans (4)**

NCERT, Pg. # 149

157. **Ans (3)**

NCERT, Pg. # 156

158. **Ans (4)**

NCERT, Pg. # 157

159. **Ans (4)**

NCERT, Pg. # 146

160. **Ans (4)**

NCERT, Pg. # 92

161. **Ans (2)**

NCERT, Pg. # 156

162. **Ans (4)**

NCERT, Pg. # 159

163. **Ans (3)**

NCERT, Pg. # 153

164. **Ans (4)**

NCERT, Pg. # 152

165. **Ans (3)**

NCERT, Pg. # 147

166. **Ans (3)**

NCERT, Pg. # 158

167. **Ans (4)**

NCERT, Pg. # 154

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168. **Ans (3)**

NCERT, Pg. # 153

173. **Ans (3)**

NCERT XII, Pg # 85

SECTION-B

189. **Ans (4)**

NCERT (XIIth) Pg.# 160, (Para 8.5)

190. **Ans (1)**

NCERT-XII Pg. No. 151

191. **Ans (2)**

NCERT XII Pg. # 157

192. **Ans (2)**

NCERT, Pg. # 147

193. **Ans (3)**

NCERT Pg. # 78 (concept)

194. **Ans (4)**

NCERT-XII Pg.# 78

195. **Ans (4)**

NCERT-XII Pg. # 83 (E)

NCERT-XII Pg. # 93 (H)

196. **Ans (3)**

NCERT-XII, Page# 84

199. **Ans (4)**

NCERT-XII Pg. # 89